2.53 Polar Integration

Worked Solutions

1(a)

$$\int (2\cos 3x)^2\,dx = \int 4\cos^2(3x)\,dx.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int 4\cos^2(3x)\,dx &= \int (2 + 2\cos 6x)\,dx \\ &= 2x + \frac{1}{3}\sin 6x + c.\end{aligned}$$

1(b)

$$\int (2\sin 3x)^2\,dx = \int 4\sin^2(3x)\,dx.$$

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int 4\sin^2(3x)\,dx &= \int (2 - 2\cos 6x)\,dx \\ &= 2x - \frac{1}{3}\sin 6x + c.\end{aligned}$$

1(c)

$$\int (1+\cos x)^2\,dx = \int (1 + 2\cos x + \cos^2 x)\,dx.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int (1 + 2\cos x + \cos^2 x)\,dx &= \int \left(\frac{3}{2} + 2\cos x + \frac{1}{2}\cos 2x\right) dx \\ &= \frac{3}{2}x + 2\sin x + \frac{1}{4}\sin 2x + c.\end{aligned}$$

1(d)

$$\int (1-\sin x)^2\,dx = \int (1 - 2\sin x + \sin^2 x)\,dx.$$

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int (1 - 2\sin x + \sin^2 x)\,dx &= \int \left(\frac{3}{2} - 2\sin x - \frac{1}{2}\cos 2x\right) dx \\ &= \frac{3}{2}x + 2\cos x - \frac{1}{4}\sin 2x + c.\end{aligned}$$

1(e)

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int \sin^2\left(\frac{x}{6}\right) dx &= \int \left(\frac{1}{2} - \frac{1}{2}\cos\frac{x}{3}\right) dx \\ &= \frac{1}{2}x - \frac{3}{2}\sin\frac{x}{3} + c.\end{aligned}$$

1(f)

$$\int \left(2 + \cos\frac{x}{2}\right)^2 dx = \int \left(4 + 4\cos\frac{x}{2} + \cos^2\frac{x}{2}\right) dx.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\int \left(4 + 4\cos\frac{x}{2} + \cos^2\frac{x}{2}\right) dx &= \int \left(\frac{9}{2} + 4\cos\frac{x}{2} + \frac{1}{2}\cos x\right) dx \\ &= \frac{9}{2}x + 8\sin\frac{x}{2} + \frac{1}{2}\sin x + c.\end{aligned}$$

2(a)

$$\frac{1}{2}\int (4 + 3\cos\theta)^2 d\theta = \frac{1}{2}\int (16 + 24\cos\theta + 9\cos^2\theta) d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}\int (16 + 24\cos\theta + 9\cos^2\theta) d\theta &= \frac{1}{2}\int \left(\frac{41}{2} + 24\cos\theta + \frac{9}{2}\cos 2\theta\right) d\theta \\ &= \frac{1}{2}\left(\frac{41}{2}\theta + 24\sin\theta + \frac{9}{4}\sin 2\theta\right) + c.\end{aligned}$$

$$\begin{aligned}\frac{1}{2}\int_0^{2\pi} (4 + 3\cos\theta)^2 d\theta &= \left[\frac{1}{2}\left(\frac{41}{2}\theta + 24\sin\theta + \frac{9}{4}\sin 2\theta\right)\right]_0^{2\pi} \\ &= \frac{1}{2}(41\pi + 0 + 0) - \frac{1}{2}(0 + 0 + 0) \\ &= \frac{41}{2}\pi \text{ units}^2.\end{aligned}$$

2(b)

$$\frac{1}{2}\int (5+2\sin\theta)^2 d\theta = \frac{1}{2}\int (25+20\sin\theta+4\sin^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1-2\sin^2 A, \quad \sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}\int (25+20\sin\theta+4\sin^2\theta)d\theta &= \frac{1}{2}\int (27+20\sin\theta-2\cos 2\theta)d\theta \\
&= \frac{1}{2}(27\theta-20\cos\theta-\sin 2\theta)+c. \\
\frac{1}{2}\int_0^{2\pi} (5+2\sin\theta)^2 d\theta &= \left[\frac{1}{2}(27\theta-20\cos\theta-\sin 2\theta)\right]_0^{2\pi} \\
&= \frac{1}{2}(54\pi-20-0)-\frac{1}{2}(0-20-0) \\
&= 27\pi \text{ units}^2.
\end{aligned}$$

2(c)

$$\frac{1}{2}\int \left(a(1-\cos\theta)\right)^2 d\theta = \frac{1}{2}a^2\int (1-2\cos\theta+\cos^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A-1, \quad \cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}a^2\int (1-2\cos\theta+\cos^2\theta)d\theta &= \frac{1}{2}a^2\int \left(\frac{3}{2}-2\cos\theta+\frac{1}{2}\cos 2\theta\right)d\theta \\
&= \frac{1}{2}a^2\left(\frac{3}{2}\theta-2\sin\theta+\frac{1}{4}\sin 2\theta\right)+c. \\
\frac{1}{2}\int_0^{2\pi} \left(a(1-\cos\theta)\right)^2 d\theta &= \left[\frac{1}{2}a^2\left(\frac{3}{2}\theta-2\sin\theta+\frac{1}{4}\sin 2\theta\right)\right]_0^{2\pi} \\
&= \frac{1}{2}a^2\left(\frac{3}{2}\cdot 2\pi-0+0\right)-\frac{1}{2}a^2(0-0+0) \\
&= \frac{3}{2}a^2\pi \text{ units}^2.
\end{aligned}$$

2(d)

$$\frac{1}{2}\int \left(a(1-\sin\theta)\right)^2 d\theta = \frac{1}{2}a^2\int (1-2\sin\theta+\sin^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1-2\sin^2 A, \quad \sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}a^2\int (1-2\sin\theta+\sin^2\theta)d\theta &= \frac{1}{2}a^2\int \left(\frac{3}{2}-2\sin\theta-\frac{1}{2}\cos 2\theta\right)d\theta \\
&= \frac{1}{2}a^2\left(\frac{3}{2}\theta+2\cos\theta-\frac{1}{4}\sin 2\theta\right)+c. \\
\frac{1}{2}\int_0^{2\pi} \left(a(1-\sin\theta)\right)^2 d\theta &= \left[\frac{1}{2}a^2\left(\frac{3}{2}\theta+2\cos\theta-\frac{1}{4}\sin 2\theta\right)\right]_0^{2\pi} \\
&= \frac{1}{2}a^2\left(3\pi+2-0\right)-\frac{1}{2}a^2(0+2-0) \\
&= \frac{3}{2}a^2\pi \text{ units}^2.
\end{aligned}$$

3(a)

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}\int (a\cos\theta)^2 d\theta &= \frac{1}{2}a^2 \int \left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right) d\theta \\
&= \frac{1}{2}a^2 \left(\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right) + c. \\
\frac{1}{2}\int_{-\pi/2}^{\pi/2} (a\cos\theta)^2 d\theta &= \left[\frac{1}{2}a^2 \left(\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right)\right]_{-\pi/2}^{\pi/2} \\
&= \frac{1}{2}a^2 \left(\frac{\pi}{4} + 0\right) - \frac{1}{2}a^2 \left(-\frac{\pi}{4} + 0\right) \\
&= \frac{1}{4}a^2\pi \text{ units}^2.
\end{aligned}$$

3(b)

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}\int (a\sin\theta)^2 d\theta &= \frac{1}{2}a^2 \int \left(\frac{1}{2} - \frac{1}{2}\cos 2\theta\right) d\theta \\
&= \frac{1}{2}a^2 \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right) + c. \\
\frac{1}{2}\int_0^{\pi} (a\sin\theta)^2 d\theta &= \left[\frac{1}{2}a^2 \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right)\right]_0^{\pi} \\
&= \frac{1}{2}a^2 \left(\frac{1}{2}\pi - 0\right) - \frac{1}{2}a^2(0 - 0) \\
&= \frac{1}{4}a^2\pi \text{ units}^2.
\end{aligned}$$

3(c)

$$\begin{aligned}
\frac{1}{2}\int a^2 d\theta &= \frac{1}{2}a^2 \int 1\, d\theta = \frac{1}{2}a^2\theta + c. \\
\frac{1}{2}\int_0^{2\pi} a^2 d\theta &= \left[\frac{1}{2}a^2\theta\right]_0^{2\pi} = a^2\pi \text{ units}^2.
\end{aligned}$$

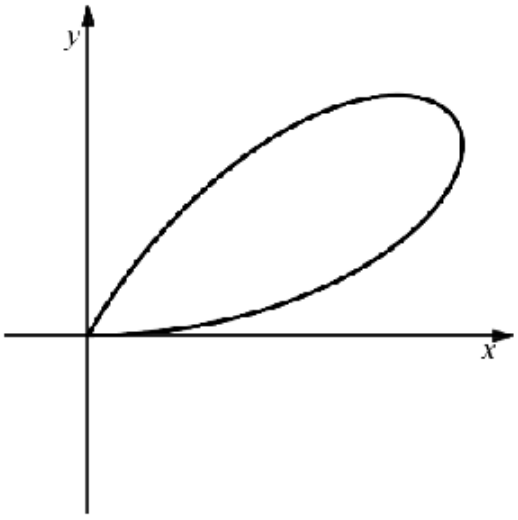
4(a)

$$\begin{aligned}
\frac{1}{2}\int_0^{\pi} (ae^{\theta})^2 d\theta &= \frac{1}{2}a^2 \int_0^{\pi} e^{2\theta} d\theta \\
&= \left[\frac{1}{4}a^2 e^{2\theta}\right]_0^{\pi} \\
&= \frac{1}{4}a^2 e^{2\pi} - \frac{1}{4}a^2 \\
&= \frac{1}{4}a^2(e^{2\pi} - 1) \text{ units}^2.
\end{aligned}$$

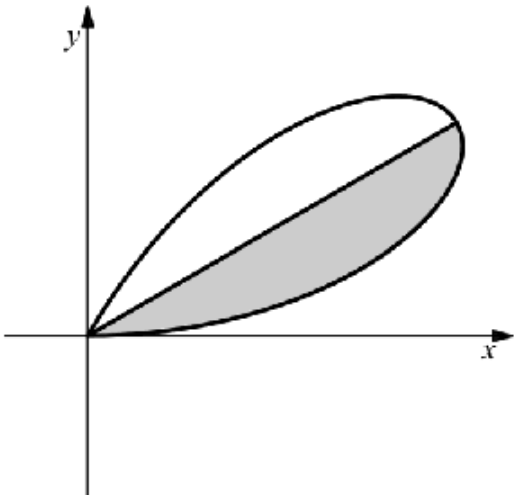
4(b)

$$\begin{aligned}\frac{1}{2}\int_0^{\pi}(a\theta)^2 d\theta &= \frac{1}{2}a^2\int_0^{\pi}\theta^2 d\theta \\ &= \left[\frac{1}{6}a^2\theta^3\right]_0^{\pi} \\ &= \frac{1}{6}a^2\pi^3 \\ &= \frac{1}{6}a^2\pi^3 \text{ units}^2.\end{aligned}$$

5(a)



5(b)



5(c)

$$\frac{1}{2}\int_0^{\pi/6}(2\cos 3\theta)^2 d\theta = 2\int_0^{\pi/6}\cos^2(3\theta)d\theta.$$

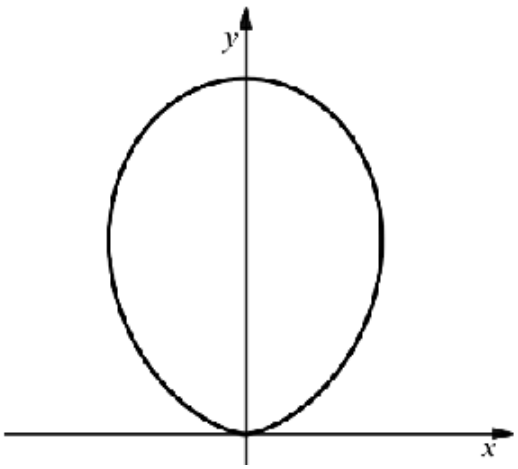
Since $\cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$

$$\begin{aligned} 2\int_0^{\pi/6}\cos^2(3\theta)d\theta &= 2\int_0^{\pi/6}\left(\frac{1}{2} + \frac{1}{2}\cos 6\theta\right)d\theta \\ &= \int_0^{\pi/6}(1+\cos 6\theta)d\theta \\ &= \left[\theta + \frac{1}{6}\sin 6\theta\right]_0^{\pi/6} \\ &= \left(\frac{\pi}{6} + 0\right) - (0+0) = \frac{\pi}{6} \text{ units}^2. \end{aligned}$$

6(a)

When $\theta = 0$, $r = 0$, and when $\theta = \pi$, $r = 0$, so the curve starts and ends at the pole. For $0 < \theta < \pi$, $\sin\theta$ increases to a maximum before decreasing, so $\sin^2\theta$ also increases to a maximum before decreasing. Therefore the curve $r = a\sin^2\theta$ for $0 \le \theta \le \pi$, for positive constant a, consists of a single closed loop through the pole.

6(b)



6(c)

$$\frac{1}{2}\int_0^{\pi}(a\sin^2\theta)^2 d\theta = \frac{1}{2}a^2\int_0^{\pi}\sin^4\theta\, d\theta.$$

Since $\cos(2A) = 1 - 2\sin^2 A$, $\quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$.

$$\begin{aligned}\sin^4 A &= \left(\frac{1}{2} - \frac{1}{2}\cos 2A\right)^2 \\ &= \frac{1}{4} - \frac{1}{2}\cos 2A + \frac{1}{4}\cos^2 2A.\end{aligned}$$

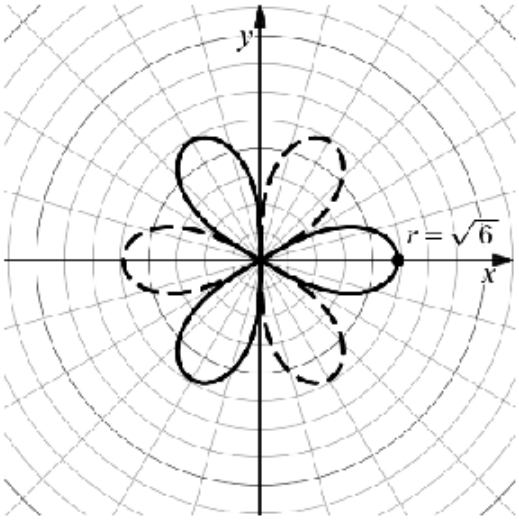
Since $\cos(2A) = 2\cos^2 A - 1$, $\quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$.

$$\begin{aligned}\sin^4 A &= \frac{1}{4} - \frac{1}{2}\cos 2A + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos 4A\right) \\ &= \frac{3}{8} - \frac{1}{2}\cos 2A + \frac{1}{8}\cos 4A.\end{aligned}$$

$$\begin{aligned}\frac{1}{2}a^2\int_0^{\pi}\sin^4\theta\, d\theta &= \frac{1}{2}a^2\int_0^{\pi}\left(\frac{3}{8} - \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta\right) d\theta \\ &= \left[\frac{1}{2}a^2\left(\frac{3}{8}\theta - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta\right)\right]_0^{\pi} \\ &= \frac{1}{2}a^2\left(\frac{3}{8}\pi - 0 + 0\right) - \frac{1}{2}a^2(0 - 0 + 0) \\ &= \frac{3}{16}\pi a^2 \text{ units}^2.\end{aligned}$$

7(a)

Since $r^2 = 6\cos 3\theta$, consider $r = \pm\sqrt{6\cos 3\theta}$ and use dashed lines for negative values of r.



7(b)

The student has found the sum of both the positive contributions and the negative contributions, so they have cancelled out to form the answer zero.

7(c)

$$\begin{aligned}\frac{1}{2}\int_{-\pi/6}^{\pi/6} 6\cos 3\theta\, d\theta &= \left[\frac{1}{2}(2\sin 3\theta)\right]_{-\pi/6}^{\pi/6} \\ &= \left(\frac{1}{2}\times 2\right) - \left(\frac{1}{2}\times(-2)\right) \\ &= 2 \text{ units}^2.\end{aligned}$$

7(d)

Total area is 12 units^2.

8(a)

$$\sin 2x = 0 \text{ at } x = 0 \text{ and } x = \frac{\pi}{2}.$$

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/2} a\sin 2\theta\, d\theta &= \left[-\frac{1}{4}a\cos 2\theta\right]_0^{\pi/2} \\ &= \frac{1}{4}a - \left(-\frac{1}{4}a\right) = \frac{1}{2}a.\end{aligned}$$

So two loops gives: $2\times\frac{1}{2}a = a$ units^2.

8(b)

$$\cos 2x = 0 \text{ at } x = -\frac{\pi}{4} \text{ and } x = \frac{\pi}{4}.$$

$$\begin{aligned}\frac{1}{2}\int_{-\pi/4}^{\pi/4} a\cos 2\theta\, d\theta &= \left[\frac{1}{4}a\sin 2\theta\right]_{-\pi/4}^{\pi/4} \\ &= \frac{1}{4}a - \left(-\frac{1}{4}a\right) = \frac{1}{2}a.\end{aligned}$$

So two loops gives: $2\times\frac{1}{2}a = a$ units^2.

8(c)

$$\sin 3x = 0 \text{ at } x = 0 \text{ and } x = \frac{\pi}{3}.$$

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/3} a\sin 3\theta\, d\theta &= \left[-\frac{1}{6}a\cos 3\theta\right]_0^{\pi/3} \\ &= \frac{1}{6}a - \left(-\frac{1}{6}a\right) = \frac{1}{3}a.\end{aligned}$$

So six loops gives: $6\times\frac{1}{3}a = 2a$ units^2.

8(d)

$$\cos 3x = 0 \text{ at } x = -\frac{\pi}{6} \text{ and } x = \frac{\pi}{6}.$$

$$\begin{aligned}\frac{1}{2}\int_{-\pi/6}^{\pi/6} a\cos 3\theta\, d\theta &= \left[\frac{1}{6}a\sin 3\theta\right]_{-\pi/6}^{\pi/6} \\ &= \frac{1}{6}a - \left(-\frac{1}{6}a\right) = \frac{1}{3}a.\end{aligned}$$

So six loops gives: $6 \times \frac{1}{3}a = 2a$ units^2.

9(a)

$$\cos 3x = 0 \text{ at } x = -\frac{\pi}{6} \text{ and } x = \frac{\pi}{6}.$$

$$\frac{1}{2}\int_{-\pi/6}^{\pi/6}(2\cos 3\theta)^2 d\theta = 2\int_{-\pi/6}^{\pi/6}\cos^2(3\theta)d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}2\int_{-\pi/6}^{\pi/6}\cos^2(3\theta)d\theta &= \int_{-\pi/6}^{\pi/6}(1+\cos 6\theta)d\theta \\ &= \left[\theta + \frac{1}{6}\sin 6\theta\right]_{-\pi/6}^{\pi/6} \\ &= \left(\frac{\pi}{6}+0\right) - \left(-\frac{\pi}{6}+0\right) = \frac{\pi}{3}.\end{aligned}$$

So three loops gives: $3 \times \frac{\pi}{3} = \pi$ units^2.

9(b)

$$\sin 2x = 0 \text{ at } x = 0 \text{ and } x = \frac{\pi}{2}.$$

$$\frac{1}{2}\int_{0}^{\pi/2}(3\sin 2\theta)^2 d\theta = \frac{9}{2}\int_{0}^{\pi/2}\sin^2(2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{9}{2}\int_{0}^{\pi/2}\sin^2(2\theta)d\theta &= \frac{9}{4}\int_{0}^{\pi/2}(1-\cos 4\theta)d\theta \\ &= \left[\frac{9}{4}\left(\theta - \frac{1}{4}\sin 4\theta\right)\right]_{0}^{\pi/2} \\ &= \frac{9}{4}\left(\frac{\pi}{2}-0\right) - (0-0) = \frac{9\pi}{8}.\end{aligned}$$

So four loops gives: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$ units^2.

9(c)

$$\cos 4x = 0 \text{ at } x = -\frac{\pi}{8} \text{ and } x = \frac{\pi}{8}.$$

$$\frac{1}{2}\int_{-\pi/8}^{\pi/8} (a\cos 4\theta)^2 d\theta = \frac{1}{2}a^2 \int_{-\pi/8}^{\pi/8} \cos^2(4\theta) d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}a^2 \int_{-\pi/8}^{\pi/8} \cos^2(4\theta) d\theta &= \frac{1}{4}a^2 \int_{-\pi/8}^{\pi/8} (1 + \cos 8\theta) d\theta \\ &= \left[\frac{1}{4}a^2\left(\theta + \frac{1}{8}\sin 8\theta\right)\right]_{-\pi/8}^{\pi/8} \\ &= \frac{1}{4}a^2\left(\frac{\pi}{8} + 0\right) - \frac{1}{4}a^2\left(-\frac{\pi}{8} + 0\right) = \frac{\pi a^2}{16}.\end{aligned}$$

So eight loops gives: $8 \times \dfrac{\pi a^2}{16} = \dfrac{1}{2}\pi a^2$ units^2.

9(d)

$$\sin 5x = 0 \text{ at } x = 0 \text{ and } x = \frac{\pi}{5}.$$

$$\frac{1}{2}\int_{0}^{\pi/5} (a\sin 5\theta)^2 d\theta = \frac{1}{2}a^2 \int_{0}^{\pi/5} \sin^2(5\theta) d\theta.$$

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}a^2 \int_{0}^{\pi/5} \sin^2(5\theta) d\theta &= \frac{1}{4}a^2 \int_{0}^{\pi/5} (1 - \cos 10\theta) d\theta \\ &= \left[\frac{1}{4}a^2\left(\theta - \frac{1}{10}\sin 10\theta\right)\right]_{0}^{\pi/5} \\ &= \frac{1}{4}a^2\left(\frac{\pi}{5} - 0\right) - (0 - 0) = \frac{\pi a^2}{20}.\end{aligned}$$

So five loops gives: $5 \times \dfrac{\pi a^2}{20} = \dfrac{1}{4}\pi a^2$ units^2.

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$$\begin{aligned}\frac{1}{2}\int_{\pi/6}^{11\pi/6} \left(\frac{1}{\sqrt{\theta}}\right)^2 d\theta &= \frac{1}{2}\int_{\pi/6}^{11\pi/6} \frac{1}{\theta}\, d\theta \\ &= \left[\frac{1}{2}\ln\theta\right]_{\pi/6}^{11\pi/6} \\ &= \frac{1}{2}\ln\frac{11\pi}{6} - \frac{1}{2}\ln\frac{\pi}{6} \\ &= \frac{1}{2}\ln 11 \text{ units}^2.\end{aligned}$$

11(a)

$$0 = 1 + 2\cos\theta \quad \Longrightarrow \quad \theta = \frac{2\pi}{3},\ \frac{4\pi}{3}.$$

$$\text{So } r < 0 \text{ for } \frac{2\pi}{3} < \theta < \frac{4\pi}{3}.$$

$$\frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1+2\cos\theta)^2 d\theta = \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1+4\cos\theta+4\cos^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1+4\cos\theta+4\cos^2\theta)d\theta &= \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(3+4\cos\theta+2\cos 2\theta)d\theta \\ &= \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{2}\sin 2\theta\right]_{2\pi/3}^{4\pi/3} \\ &= \left(2\pi - \sqrt{3} + \frac{\sqrt{3}}{4}\right) - \left(\pi + \sqrt{3} - \frac{\sqrt{3}}{4}\right) \\ &= \left(\pi - \frac{3\sqrt{3}}{2}\right) \text{ units}^2.\end{aligned}$$

11(b)

$$0 = 2 - 4\sin\theta \quad \Longrightarrow \quad \theta = \frac{\pi}{6},\ \frac{5\pi}{6}.$$

$$\text{So } r < 0 \text{ for } \frac{\pi}{6} < \theta < \frac{5\pi}{6}.$$

$$\frac{1}{2}\int_{\pi/6}^{5\pi/6}(2-4\sin\theta)^2 d\theta = \frac{1}{2}\int_{\pi/6}^{5\pi/6}(4-16\sin\theta+16\sin^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1 - 2\sin^2 A, \quad \sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}\int_{\pi/6}^{5\pi/6}(4-16\sin\theta+16\sin^2\theta)d\theta &= \int_{\pi/6}^{5\pi/6}(6-8\sin\theta-4\cos 2\theta)d\theta \\ &= [6\theta + 8\cos\theta - 2\sin 2\theta]_{\pi/6}^{5\pi/6} \\ &= \left(5\pi - 4\sqrt{3} + \sqrt{3}\right) - \left(\pi + 4\sqrt{3} - \sqrt{3}\right) \\ &= (4\pi - 6\sqrt{3}) \text{ units}^2.\end{aligned}$$

12

$$\frac{1}{2}\int_0^{2\pi}\left(a(p+q\sin\theta)\right)^2 d\theta = \frac{1}{2}a^2\int_0^{2\pi}(p^2+2pq\sin\theta+q^2\sin\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1-2\sin^2 A, \quad \sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A).$$

$$\begin{aligned}
\frac{1}{2}a^2\int_0^{2\pi}(p^2+2pq\sin\theta+q^2\sin^2\theta)d\theta &= \frac{1}{2}a^2\int_0^{2\pi}\left(p^2+2pq\sin\theta+\frac{1}{2}q^2-\frac{1}{2}q^2\cos 2\theta\right)d\theta \\
&= \left[\frac{1}{2}a^2\left(\left(p^2+\frac{1}{2}q^2\right)\theta-2pq\cos\theta-\frac{1}{4}q^2\sin 2\theta\right)\right]_0^{2\pi} \\
&= \frac{1}{2}a^2\left(\left(p^2+\frac{1}{2}q^2\right)2\pi-2pq-0\right)-\frac{1}{2}a^2(0-2pq-0) \\
&= 2\pi a^2\left(\frac{1}{2}p^2+\frac{1}{4}q^2\right) \\
&= \frac{1}{2}\pi a^2(2p^2+q^2)\text{ units}^2 \quad \text{as required.}
\end{aligned}$$

13(a)

The curve is symmetrical across the initial line, so the ratio of the areas is $1:1$.

13(b)

$$\frac{1}{2}\int_0^{\pi}\left(2(1+\sin\theta)\right)^2 d\theta = 2\int_0^{\pi}(1+2\sin\theta+\sin^2\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1-2\sin^2 A, \quad \sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A).$$

$$\begin{aligned}
2\int_0^{\pi}(1+2\sin\theta+\sin^2\theta)d\theta &= 2\int_0^{\pi}\left(\frac{3}{2}+2\sin\theta-\frac{1}{2}\cos 2\theta\right)d\theta \\
&= \left[2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin 2\theta\right)\right]_0^{\pi} \\
&= 2\left(\frac{3}{2}\pi+2-0\right)-2(0-2-0) = 8+3\pi.
\end{aligned}$$

$$\begin{aligned}
\frac{1}{2}\int_{\pi}^{2\pi}\left(2(1+\sin\theta)\right)^2 d\theta &= 2\int_{\pi}^{2\pi}(1+2\sin\theta+\sin^2\theta)d\theta \\
&= \left[2\left(\frac{3}{2}\theta-2\cos\theta-\frac{1}{4}\sin 2\theta\right)\right]_{\pi}^{2\pi} \\
&= 2(3\pi-2-0)-2\left(\frac{3}{2}\pi+2-0\right) = -8+3\pi.
\end{aligned}$$

$$\therefore \quad \text{ratio} = (8+3\pi):(-8+3\pi).$$

14(a)

$$\frac{1}{2}\int_0^{2\pi}\left(\frac{2}{5}(4+\sin 4\theta)\right)^2 d\theta = \frac{2}{25}\int_0^{2\pi}(16+8\sin 4\theta+\sin^2 4\theta)d\theta.$$

$$\text{Since } \cos(2A) = 1-2\sin^2 A, \quad \sin^2 A = \frac{1}{2}-\frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{2}{25}\int_0^{2\pi}(16+8\sin 4\theta+\sin^2 4\theta)d\theta &= \frac{2}{25}\int_0^{2\pi}\left(\frac{33}{2}+8\sin 4\theta-\frac{1}{2}\cos 8\theta\right)d\theta\\ &= \left[\frac{2}{25}\left(\frac{33}{2}\theta-2\cos 4\theta-\frac{1}{16}\sin 8\theta\right)\right]_0^{2\pi}\\ &= \frac{2}{25}\left(\frac{33}{2}\cdot 2\pi-2-0\right)-\frac{2}{25}(0-2-0)\\ &= \frac{66}{25}\pi.\end{aligned}$$

$$\text{Volume} = 0.8\times\frac{66}{25}\pi = \frac{264}{125}\pi \text{ m}^3.$$

14(b)

All the water stays in the pond; there is no evaporation or overflow. The pond is filled exactly to the top. The pond exactly matches the curve.

15(a)

$$\frac{1}{2}\int_0^{2\pi}(5+\cos 4\theta)^2 d\theta = \frac{1}{2}\int_0^{2\pi}(25+10\cos 4\theta+\cos^2 4\theta)d\theta.$$

$$\text{Since } \cos(2A) = 2\cos^2 A-1, \quad \cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A).$$

$$\begin{aligned}\frac{1}{2}\int_0^{2\pi}(25+10\cos 4\theta+\cos^2 4\theta)d\theta &= \int_0^{2\pi}\left(\frac{51}{4}+5\cos 4\theta+\frac{1}{4}\cos 8\theta\right)d\theta\\ &= \left[\frac{51}{4}\theta+\frac{5}{4}\sin 4\theta+\frac{1}{32}\sin 8\theta\right]_0^{2\pi}\\ &= \frac{51}{4}\cdot 2\pi = \frac{51}{2}\pi.\end{aligned}$$

$$\text{Volume} = 0.4\times\frac{51}{2}\pi = \frac{51}{5}\pi \text{ m}^3.$$

$$40 \text{ litres per minute} = 0.04 \text{ m}^3 \text{ per minute.}$$

$$\text{Time} = \frac{(51/5)\pi}{0.04} \approx 801 \text{ minutes.}$$

15(b)

The pool is assumed to have vertical sides at every point. The pool has uniform depth. The pool is filled exactly to the top (not less than the full amount or with overspill). The flow rate of the water is constant as claimed.

16(a)

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/4}(ae^{k\theta})^2 d\theta &= \frac{1}{2}a^2\int_0^{\pi/4} e^{2k\theta}d\theta \\ &= \left[\frac{1}{2}a^2\left(\frac{1}{2k}e^{2k\theta}\right)\right]_0^{\pi/4} \\ &= \frac{a^2}{4k}\left(e^{\pi k/2}-1\right)\ \text{units}^2.\end{aligned}$$

16(b)

$$\begin{aligned}\frac{1}{2}\int_{2\pi}^{9\pi/4}(ae^{k\theta})^2 d\theta &= \frac{1}{2}a^2\int_{2\pi}^{9\pi/4} e^{2k\theta}d\theta \\ &= \left[\frac{a^2}{4k}e^{2k\theta}\right]_{2\pi}^{9\pi/4} \\ &= \frac{a^2}{4k}\left(e^{9\pi k/2}-e^{4\pi k}\right) \\ &= \frac{a^2}{4k}e^{4\pi k}\left(e^{\pi k/2}-1\right)\ \text{units}^2.\end{aligned}$$

16(c)

$$\begin{aligned}\frac{1}{2}\int_{4\pi}^{17\pi/4}(ae^{k\theta})^2 d\theta &= \frac{1}{2}a^2\int_{4\pi}^{17\pi/4} e^{2k\theta}d\theta \\ &= \left[\frac{a^2}{4k}e^{2k\theta}\right]_{4\pi}^{17\pi/4} \\ &= \frac{a^2}{4k}\left(e^{17\pi k/2}-e^{8\pi k}\right) \\ &= \frac{a^2}{4k}e^{8\pi k}\left(e^{\pi k/2}-1\right)\ \text{units}^2.\end{aligned}$$

16(d)

For $n=0$, the area is $\frac{a^2}{4k}(e^{\pi k/2}-1)$. For $n=1$, the area is $\frac{a^2}{4k}e^{4\pi k}(e^{\pi k/2}-1)$. For $n=2$, the area is $\frac{a^2}{4k}e^{8\pi k}(e^{\pi k/2}-1)$. This is a geometric sequence with first term $\frac{a^2}{4k}(e^{\pi k/2}-1)$ and common ratio $e^{4\pi k}$.

17

$$\frac{1}{2}\int_0^{\pi}(1+\sinh\theta)^2 d\theta = \frac{1}{2}\int_0^{\pi}(1+2\sinh\theta+\sinh^2\theta)d\theta.$$

Since $\cosh(2A) = 1+2\sinh^2 A$, $\quad \sinh^2 A = \frac{1}{2}\cosh(2A) - \frac{1}{2}$.

$$\begin{aligned}
\frac{1}{2}\int_0^{\pi}(1+2\sinh\theta+\sinh^2\theta)d\theta &= \frac{1}{2}\int_0^{\pi}\left(\frac{1}{2}+2\sinh\theta+\frac{1}{2}\cosh 2\theta\right)d\theta \\
&= \left[\frac{1}{4}\theta+\cosh\theta+\frac{1}{8}\sinh 2\theta\right]_0^{\pi} \\
&= \left(\frac{1}{4}\pi+\cosh\pi+\frac{1}{8}\sinh 2\pi\right)-(0+1+0) \\
&= \frac{1}{4}\pi-1+\cosh\pi+\frac{1}{8}\sinh 2\pi \\
&= 44.8 \text{ units}^2 \text{ (3 s.f.)}.
\end{aligned}$$

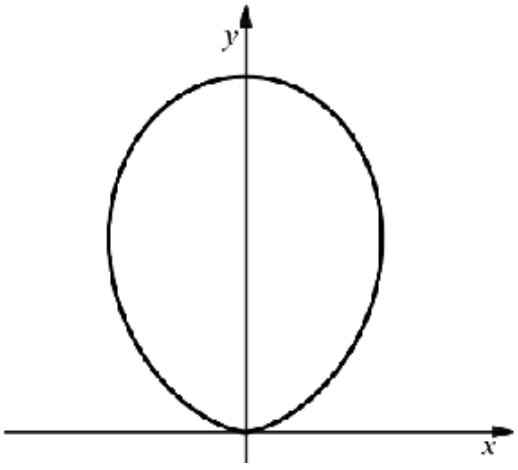
18(a)

Since $y = \sin x$ is positive for $0 < x < \pi$, $k = \pi$.

18(b)

The greatest value of r is when $\sin\theta = 1$, so $r = \sqrt{3}$.

18(c)



18(d)

$$\begin{aligned}
\frac{1}{2}\int_0^{\pi}(\sqrt{3}\sin\theta)^2 d\theta &= \frac{1}{2}\int_0^{\pi} 3\sin\theta\, d\theta \\
&= \left[-\frac{3}{2}\cos\theta\right]_0^{\pi} \\
&= \left(-\frac{3}{2}\cos\pi\right)-\left(-\frac{3}{2}\cos 0\right) = 3 \text{ units}^2.
\end{aligned}$$

19(a)

Since $y = \sin 2x$ is positive for $0 < x < \frac{\pi}{2}$, $k = \frac{\pi}{2}$.

19(b)

$$\begin{aligned}\frac{1}{2}\int_0^{\pi/2}(\sqrt{16\sin 2\theta})^2 d\theta &= 8\int_0^{\pi/2}\sin 2\theta\, d\theta \\ &= [-4\cos 2\theta]_0^{\pi/2} \\ &= (-4\cos\pi) - (-4\cos 0) = 8 \text{ units}^2.\end{aligned}$$

19(c)

r is a maximum when $\sin 2\theta = 1$, so $r = 4$. This occurs when $\theta = \frac{\pi}{4}$, since $\sqrt{16\sin(2\times\frac{\pi}{4})} = 4$. Hence $P = (4, \frac{\pi}{4})$.

19(d)

The Cartesian coordinates of P are

$$\left(4\cos\frac{\pi}{4}, 4\sin\frac{\pi}{4}\right) = (2\sqrt{2}, 2\sqrt{2}).$$

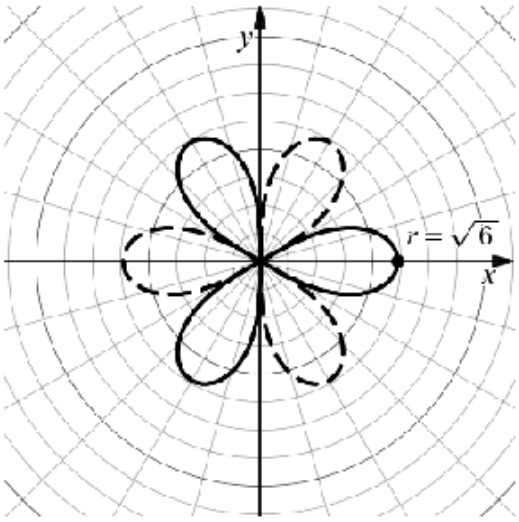
Under the transformation, the image of P is at $(8\sqrt{2}, 2\sqrt{2})$. Therefore $r\cos\theta = 8\sqrt{2}$ and $r\sin\theta = 2\sqrt{2}$, so

$$\tan\theta = \frac{2\sqrt{2}}{8\sqrt{2}} = \frac{1}{4}.$$

19(e)

$\det M = 1 - 0 = 1$, so the area remains the same: 8 units^2.

20(a)



20(b)

Inside the loop is when $\frac{2\pi}{3} \le \theta < \frac{4\pi}{3}$. The required area is

$$\begin{aligned}
A_R &= \frac{1}{2}\int_0^{2\pi/3}(2+4\cos\theta)^2 d\theta \\
&\quad + \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(2+4\cos\theta)^2 d\theta \\
&\quad - \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(2+4\cos\theta)^2 d\theta. \\
\frac{1}{2}\int(2+4\cos\theta)^2 d\theta &= \int(2+8\cos\theta+8\cos^2\theta)d\theta. \\
\text{Since } \cos(2A) &= 2\cos^2 A - 1, \quad \cos^2 A = \frac{1}{2}+\frac{1}{2}\cos(2A). \\
\int(2+8\cos\theta+8\cos^2\theta)d\theta &= \int(6+8\cos\theta+4\cos 2\theta)d\theta \\
&= 6\theta + 8\sin\theta + 2\sin 2\theta + c. \\
\frac{1}{2}\int_0^{2\pi/3}(2+4\cos\theta)^2 d\theta &= [6\theta+8\sin\theta+2\sin 2\theta]_0^{2\pi/3} \\
&= (4\pi + 4\sqrt{3} - \sqrt{3}) - (0+0+0) = 4\pi + 3\sqrt{3}. \\
\text{By symmetry, } \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(2+4\cos\theta)^2 d\theta &= 4\pi + 3\sqrt{3}. \\
\frac{1}{2}\int_{2\pi/3}^{4\pi/3}(2+4\cos\theta)^2 d\theta &= [6\theta+8\sin\theta+2\sin 2\theta]_{2\pi/3}^{4\pi/3} \\
&= (8\pi - 4\sqrt{3} + \sqrt{3}) - (4\pi + 4\sqrt{3} - \sqrt{3}) = 4\pi - 6\sqrt{3}. \\
\therefore \quad \text{required area} &= 2(4\pi + 3\sqrt{3}) - (4\pi - 6\sqrt{3}) \\
&= 4\pi + 12\sqrt{3} \text{ units}^2 \quad \text{as required.}
\end{aligned}$$

21

Using $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$,

$$\begin{aligned}
\frac{1}{2}\int\left(2-\sin\frac{\theta}{4}\right)^2 d\theta &= \frac{1}{2}\int\left(4 - 4\sin\frac{\theta}{4} + \sin^2\frac{\theta}{4}\right)d\theta \\
&= \int\left(\frac{9}{4} - 2\sin\frac{\theta}{4} - \frac{1}{4}\cos\frac{\theta}{2}\right)d\theta \\
&= \frac{9}{4}\theta + 8\cos\frac{\theta}{4} - \frac{1}{2}\sin\frac{\theta}{2} + c.
\end{aligned}$$

The three areas used in the shaded region are

$$\begin{aligned}
A_1 &= \left[\frac{9}{4}\theta + 8\cos\frac{\theta}{4} - \frac{1}{2}\sin\frac{\theta}{2}\right]_0^{\pi} = \frac{9\pi}{4} + 4\sqrt{2} - \frac{1}{2} - 8, \\
A_2 &= \left[\frac{9}{4}\theta + 8\cos\frac{\theta}{4} - \frac{1}{2}\sin\frac{\theta}{2}\right]_{\pi}^{3\pi} = \frac{9\pi}{2} - 8\sqrt{2} + 1, \\
A_3 &= A_1.
\end{aligned}$$

Hence the shaded area is $A_1 - A_2 + A_3 = 2A_1 - A_2 = -18 + 16\sqrt{2}$ units^2.

22(a)

By De Moivre's theorem,

$$z^n = \cos n\theta + i \sin n\theta, \qquad z^{-n} = \cos n\theta - i \sin n\theta,$$

so $z^n + \dfrac{1}{z^n} = 2 \cos n\theta$.

22(b)

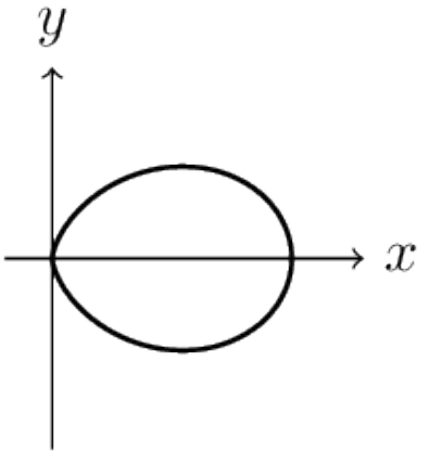
$$\begin{aligned}\left(z + \frac{1}{z}\right)^4 &= z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4} \\ &= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6.\end{aligned}$$

Using part (a), $z + \dfrac{1}{z} = 2\cos\theta$, $z^2 + \dfrac{1}{z^2} = 2\cos 2\theta$ and $z^4 + \dfrac{1}{z^4} = 2\cos 4\theta$. Therefore

$$16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6,$$

which gives $\cos^4\theta = \dfrac{1}{8}\cos 4\theta + \dfrac{1}{2}\cos 2\theta + \dfrac{3}{8}$.

22(c)



22(d)

$$\begin{aligned}A &= \frac{1}{2}\int_{-\pi/2}^{\pi/2} (a\cos^2\theta)^2\, d\theta = \frac{a^2}{2}\int_{-\pi/2}^{\pi/2} \cos^4\theta\, d\theta \\ &= \frac{a^2}{2}\int_{-\pi/2}^{\pi/2} \left(\frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}\right) d\theta \\ &= \left[\frac{a^2}{2}\left(\frac{1}{32}\sin 4\theta + \frac{1}{4}\sin 2\theta + \frac{3}{8}\theta\right)\right]_{-\pi/2}^{\pi/2} = \frac{3\pi a^2}{16}.\end{aligned}$$

23(a)

$$\begin{aligned}(x^2+y^2)^3 = y^4 &\Longrightarrow (r^2\cos^2\theta + r^2\sin^2\theta)^3 = r^4\sin^4\theta \\ &\Longrightarrow r^6 = r^4\sin^4\theta \\ &\Longrightarrow r^2 = \sin^4\theta.\end{aligned}$$

23(b)

Since $r^2 = \sin^4\theta$, take $r = \pm\sin^2\theta$. The negative values of r are represented by dashed parts of the polar curve.

23(c)

Using $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$,

$$\begin{aligned}A_{\text{one loop}} &= \frac{1}{2}\int_0^\pi \sin^4\theta\, d\theta \\ &= \frac{1}{2}\int_0^\pi \left(\frac{3}{8} - \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta\right) d\theta \\ &= \left[\frac{1}{2}\left(\frac{3}{8}\theta - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta\right)\right]_0^\pi = \frac{3\pi}{16}.\end{aligned}$$

There are two loops, so the total area is $2 \times \frac{3\pi}{16} = \frac{3\pi}{8}$ units^2.

24(a)

$$\begin{aligned}(x^2+y^2)^2 = a^2(x^2-y^2) &\Longrightarrow (r^2\cos^2\theta + r^2\sin^2\theta)^2 = a^2r^2(\cos^2\theta - \sin^2\theta) \\ &\Longrightarrow r^4 = a^2r^2\cos 2\theta \\ &\Longrightarrow r^2 = a^2\cos 2\theta.\end{aligned}$$

24(b)

Since $r^2 = a^2\cos 2\theta$, take $r = \pm a\sqrt{\cos 2\theta}$, using dashed parts for negative r. The curve exists where $\cos 2\theta \geq 0$, namely $0 \leq \theta \leq \frac{\pi}{4}$, $\frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}$, and $\frac{7\pi}{4} \leq \theta < 2\pi$.

24(c)

$$\begin{aligned}A_{\text{one loop}} &= \frac{1}{2}a^2\int_{3\pi/4}^{5\pi/4}\cos 2\theta\, d\theta = \left[\frac{a^2}{4}\sin 2\theta\right]_{3\pi/4}^{5\pi/4} \\ &= \frac{a^2}{4}\left(\sin\frac{5\pi}{2} - \sin\frac{3\pi}{2}\right) = \frac{a^2}{2}.\end{aligned}$$

There are two loops, so the total area is a^2 units^2.

25(a)

$$\begin{aligned}y^2 &= a^2(x-y)^{4/3} - x^2, \\ r^2\sin^2\theta &= a^2(r\cos\theta - r\sin\theta)^{4/3} - r^2\cos^2\theta.\end{aligned}$$

Adding $r^2\cos^2\theta$ to both sides and raising to the power 3 gives

$$\begin{aligned} r^6(\cos^2\theta+\sin^2\theta)^3 &= a^6(r\cos\theta - r\sin\theta)^4, \\ r^6 &= a^6r^4(1-\sin 2\theta)^2, \\ r^2 &= a^6(1-\sin 2\theta)^2. \end{aligned}$$

25(b)

If $y=-x$ is a line of symmetry, then $(x,y)\mapsto(-y,-x)$. Substitution gives

$$(-x)^2 = a^2((-y)-(-x))^{4/3} - (-y)^2,$$

which simplifies to $y^2 = a^2(x-y)^{4/3} - x^2$. Hence $y=-x$ is a line of symmetry.

25(c)

Since $r^2 = a^6(1-\sin 2\theta)^2$, take $r = \pm a^3(1-\sin 2\theta)$, using dashed parts when $r<0$. The expression $1-\sin 2\theta$ is non-negative and is zero at $\theta=\frac{\pi}{4}$ and $\theta=\frac{5\pi}{4}$.

25(d)

Using $\sin^2(2\theta) = \frac{1}{2}(1-\cos 4\theta)$,

$$\begin{aligned} A_{\text{one loop}} &= \frac{1}{2}a^6\int_{\pi/4}^{5\pi/4}(1-\sin 2\theta)^2\,d\theta \\ &= \frac{1}{2}a^6\int_{\pi/4}^{5\pi/4}\left(\frac{3}{2}-2\sin 2\theta-\frac{1}{2}\cos 4\theta\right)d\theta \\ &= \left[\frac{1}{2}a^6\left(\frac{3}{2}\theta+\cos 2\theta-\frac{1}{8}\sin 4\theta\right)\right]_{\pi/4}^{5\pi/4} = \frac{3\pi a^6}{4}. \end{aligned}$$

There are two loops, so the total area is $\frac{3\pi a^6}{2}$ units^2.

26(a)

$$\begin{aligned} (x^2+y^2)^2+ax(3y^2-x^2)=0 &\Longrightarrow r^4 + ar^3\cos\theta(3\sin^2\theta-\cos^2\theta)=0 \\ &\Longrightarrow r = a\cos\theta(\cos^2\theta-3\sin^2\theta) \\ &= a\cos\theta(4\cos^2\theta-3) = a\cos 3\theta. \end{aligned}$$

26(b)

The curve $r = a\cos 3\theta$ has three loops. The loop centred on the positive x-axis is traced for $-\frac{\pi}{6}\le\theta\le\frac{\pi}{6}$; the other two loops are congruent.

26(c)

$$\begin{aligned} A_{\text{one loop}} &= \frac{1}{2}a^2\int_{-\pi/6}^{\pi/6}\cos^2(3\theta)\,d\theta \\ &= \frac{1}{2}a^2\int_{-\pi/6}^{\pi/6}\left(\frac{1}{2}+\frac{1}{2}\cos 6\theta\right)d\theta = \frac{\pi a^2}{12}. \end{aligned}$$

Thus the total area is $3\times\frac{\pi a^2}{12} = \frac{\pi a^2}{4}$ units^2.

27(a)

$$\begin{aligned} A &= \frac{1}{2}\int_0^{\pi} (3\cos\theta + \sin\theta)^2 \, d\theta \\ &= \frac{1}{2}\int_0^{\pi} (8\cos^2\theta + 3\sin 2\theta + 1) \, d\theta \\ &= \frac{1}{2}\int_0^{\pi} (5 + 4\cos 2\theta + 3\sin 2\theta) \, d\theta \\ &= \left[\frac{5}{2}\theta + \sin 2\theta - \frac{3}{4}\cos 2\theta\right]_0^{\pi} = \frac{5\pi}{2} \text{ units}^2. \end{aligned}$$

27(b)

$$\frac{dr}{d\theta} = -3\sin\theta + \cos\theta = 0 \Longrightarrow \tan\theta = \frac{1}{3}.$$

Thus $\sin\theta = \frac{1}{\sqrt{10}}$ and $\cos\theta = \frac{3}{\sqrt{10}}$. Then $r = \frac{10}{\sqrt{10}}$ and $(x, y) = (r\cos\theta, r\sin\theta) = (3, 1)$.

27(c)

$$\begin{aligned} r^2 &= 3r\cos\theta + r\sin\theta, \\ x^2 + y^2 &= 3x + y, \\ \left(x - \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 &= \frac{5}{2}. \end{aligned}$$

This is a circle with centre $\left(\frac{3}{2}, \frac{1}{2}\right)$ and radius $\sqrt{\frac{5}{2}}$.

28

$$\begin{aligned} A &= \frac{1}{2}\int_{-\pi/2}^{\pi/2} (2(1 - \sin\theta)\cos\theta)^2 \, d\theta \\ &= 2\int_{-\pi/2}^{\pi/2} \cos^2\theta \, d\theta - 4\int_{-\pi/2}^{\pi/2} \sin\theta\cos^2\theta \, d\theta + 2\int_{-\pi/2}^{\pi/2} \sin^2\theta\cos^2\theta \, d\theta. \end{aligned}$$

The three integrals are π, 0, and $\frac{\pi}{4}$, respectively. For the last, use $2\sin^2\theta\cos^2\theta = \frac{1}{2}\sin^2(2\theta)$ and $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$. Therefore $A = \pi + 0 + \frac{\pi}{4} = \frac{5\pi}{4}$ units^2.

29(a)

$r = a\sec\theta$ gives $r\cos\theta = a$, so the curve is the straight line $x = a$.

29(b)

$$A = \frac{1}{2}\int_0^k (a\sec\theta)^2 d\theta = \left[\frac{1}{2}a^2\tan\theta\right]_0^k = \frac{1}{2}a^2\tan k \text{ units}^2.$$

30

Let $u = \tan\frac{\theta}{2}$, so $du = \frac{1}{2}\sec^2\frac{\theta}{2}\,d\theta$. Then

$$\begin{aligned} A &= \frac{1}{2}a^2\int_0^{\pi/4}\sec^4\frac{\theta}{2}\,d\theta \\ &= a^2\int_0^{\tan(\pi/8)}(1+u^2)\,du \\ &= a^2\left(\tan\frac{\pi}{8}+\frac{1}{3}\tan^3\frac{\pi}{8}\right)\ \text{units}^2. \end{aligned}$$

31(a)

Let $u = \sin\theta$, so $du = \cos\theta\,d\theta$. Then

$$\begin{aligned} A &= \frac{1}{2}\int_{-\pi/2}^{\pi/2}\left(2\sqrt{\cos\theta}\,e^{\frac{1}{3}\sin\theta}\right)^2 d\theta = 2\int_{-1}^{1}e^{\frac{2}{3}u}\,du \\ &= \left[3e^{\frac{2}{3}u}\right]_{-1}^{1} = 3\left(e^{2/3}-e^{-2/3}\right)\ \text{units}^2. \end{aligned}$$

31(b)

$$r = 2(\cos\theta)^{1/2}e^{\frac{1}{3}\sin\theta}, \qquad \frac{dr}{d\theta} = e^{\frac{1}{3}\sin\theta}\left(\frac{2}{3}\cos\theta\sqrt{\cos\theta}-\frac{\sin\theta}{\sqrt{\cos\theta}}\right).$$

At a stationary value of r, $2\cos^2\theta - 3\sin\theta = 0$. Writing $s = \sin\theta$ gives $2s^2+3s-2=0$, so $s = \frac{1}{2}$ (the other root is outside $[-1,1]$). Thus $\theta = \frac{\pi}{6}$, $\cos\theta = \frac{\sqrt{3}}{2}$, and

$$r = 2\sqrt{\frac{\sqrt{3}}{2}}e^{1/6},$$

as required.

32(a)

$$\begin{aligned} A &= \frac{1}{2}\int_0^{2\pi}\left(\frac{12}{k}+k(2+\sin\theta)\right)^2 d\theta \\ &= \frac{1}{2}\int_0^{2\pi}\left(\frac{144}{k^2}+48+24\sin\theta+4k^2+4k^2\sin\theta+k^2\sin^2\theta\right)d\theta \\ &= \int_0^{2\pi}\left(\frac{72}{k^2}+24+\frac{9}{4}k^2+(2k^2+12)\sin\theta-\frac{1}{4}k^2\cos 2\theta\right)d\theta \\ &= \left(\frac{144}{k^2}+48+\frac{9}{2}k^2\right)\pi = \left(\frac{9}{2}k^2+48+\frac{144}{k^2}\right)\pi\ \text{units}^2. \end{aligned}$$

32(b)

$$A = \pi\left(\frac{144}{k^2}+48+\frac{9}{2}k^2\right), \qquad \frac{dA}{dk} = \pi\left(-\frac{288}{k^3}+9k\right).$$

Setting $\frac{dA}{dk} = 0$ gives $9k = \frac{288}{k^3}$, so $k^4 = 32$. Since $k > 0$, the required value is $k = \sqrt[4]{32}$.

33(a)

At the initial line, $\theta = 0$ and $\theta = \pi$. Since $r_1 = \cos\theta$, the points are $(1, 0)$ and the pole $(0, \frac{\pi}{2})$.

33(b)

Both equations are unchanged when θ is replaced by $-\theta$, since $\cos(-\theta) = \cos\theta$. Therefore both curves are symmetrical about the initial line.

33(c)

$$A_{C_2} = \frac{1}{2}a^2 \int_0^{2\pi} (1 + \cos\theta)^2 d\theta = \frac{1}{2}a^2 \int_0^{2\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta\right) d\theta = \frac{3}{2}a^2\pi,$$
$$A_{C_1} = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}.$$

Hence $A_R = \left(\frac{3}{2}a^2 - \frac{1}{4}\right)\pi$ units^2.

34(a)

The maximum radius is $2.1 + a = 3$ because the design is 6 units across. Thus $a = 0.9$.

34(b)

$$\begin{aligned} A_C &= \frac{1}{2}\int_0^{2\pi} (2.1 + 0.9\cos 2\theta)^2 d\theta \\ &= \frac{1}{2}\int_0^{2\pi} (4.41 + 3.78\cos 2\theta + 0.81\cos^2 2\theta) d\theta \\ &= \frac{1}{2}\int_0^{2\pi} (4.815 + 3.78\cos 2\theta + 0.405\cos 4\theta) d\theta = 4.815\pi. \end{aligned}$$

The circle has radius 3, so its area is 9π. The shaded area is $9\pi - 4.815\pi = \frac{837\pi}{200}$ units^2.

34(c)

Scaling the logo from 6 units wide to 12 m wide multiplies its area by $2^2 = 4$. Two coats require a paint area of $2 \times 4 \times \frac{837\pi}{200} = 33.48\pi$ m^2. Dividing by 40 m^2 per tin gives approximately 2.63, so 3 tins are needed.

35(a)

At P, $\theta = \frac{\pi}{2}$ and $r = 1.3$. Thus

$$1.3 = 1 + a\sin\frac{3\pi}{2} = 1 - a,$$

so $a = -0.3$.

35(b)

$$\begin{aligned} A_{\text{curve}} &= \frac{1}{2}\int_0^{2\pi} (1 - 0.3\sin 3\theta)^2 d\theta \\ &= \int_0^{2\pi} (0.5225 - 0.3\sin 3\theta - 0.0225\cos 6\theta)\, d\theta = 1.045\pi \text{ m}^2. \end{aligned}$$

The square has area $2.6^2 = 6.76$ m^2, so the remaining area is $6.76 - 1.045\pi = 3.48$ m^2 to 3 significant figures.

36(a)

At P, $\theta = \frac{\pi}{6}$ and $r = a + b\cos\frac{\pi}{6}$. The line PQ is perpendicular to the initial line and has constant x-coordinate $r\cos\frac{\pi}{6}$. Hence its polar equation is

$$r = \frac{\sqrt{3}}{2}\left(a + \frac{\sqrt{3}}{2}b\right)\sec\theta.$$

36(b)

Since $r = 2$ at $\theta = \frac{\pi}{6}$ and $b = 1$,

$$2 = a + \cos\frac{\pi}{6} \quad \Longrightarrow \quad a = 2 - \frac{\sqrt{3}}{2}.$$

36(c)

The equilateral triangle has area $\frac{1}{2}(2)(2)\sin\frac{\pi}{3} = \sqrt{3}$. The area enclosed by the curve is

$$\begin{aligned} A_C &= \frac{1}{2}\int_{-\pi}^{\pi}\left(2 - \frac{\sqrt{3}}{2} + \cos\theta\right)^2 d\theta \\ &= \frac{\pi}{4}(21 - 8\sqrt{3}). \end{aligned}$$

Therefore the percentage is $\dfrac{\sqrt{3}}{\frac{\pi}{4}(21 - 8\sqrt{3})} \times 100\% = 30.9\%$ to 3 significant figures.

37(a)

At $\theta = \frac{\pi}{4}$, $r = 1 + \tan\frac{\pi}{4} = 2$, so $OQ = 2$ units.

37(b)

The triangle OPQ has area $\frac{1}{2}(2)(3)\sin\frac{\pi}{4} = \frac{3\sqrt{2}}{2}$. The area under the curve from 0 to $\frac{\pi}{4}$ is

$$\begin{aligned} A_C &= \frac{1}{2}\int_0^{\pi/4}(1 + \tan\theta)^2 d\theta \\ &= \int_0^{\pi/4}\left(\tan\theta + \frac{1}{2}\sec^2\theta\right)d\theta \\ &= \left[\ln|\sec\theta| + \frac{1}{2}\tan\theta\right]_0^{\pi/4} = \frac{1}{2}\ln 2 + \frac{1}{2}. \end{aligned}$$

Thus the shaded area is $\frac{3\sqrt{2}}{2} - \left(\frac{1}{2}\ln 2 + \frac{1}{2}\right) = \frac{1}{2}(3\sqrt{2} - 1 - \ln 2)$ units^2.

38(a)

C_1 is a cardioid with its cusp at the pole and symmetry about the vertical axis. C_2 is a circle with centre $(0, \frac{3}{2})$ and radius $\frac{3}{2}$.

38(b)

$$1 + \sin\theta = 3\sin\theta \Longrightarrow \sin\theta = \frac{1}{2},$$

so $\theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. In either case $r = \frac{3}{2}$, giving $(\frac{3}{2}, \frac{\pi}{6})$ and $(\frac{3}{2}, \frac{5\pi}{6})$.

38(c)

The source solutions state that the two curves do not intersect at the pole: C_1 reaches it at $\theta = \frac{3\pi}{2}$, whereas C_2 reaches it at $\theta = 0$ (or π).

39(a)

C_1 is a cardioid with cusp at the pole, symmetric about the initial line. C_2 is a circle with centre $(\frac{1}{2}, 0)$ and radius $\frac{1}{2}$.

39(b)

$$2(1 - \cos\theta) = 2\cos\theta \Longrightarrow \cos\theta = \frac{1}{2},$$

so $\theta = \pm\frac{\pi}{3}$. For each, $r = 1$, giving $(1, \frac{\pi}{3})$ and $(1, -\frac{\pi}{3})$.

39(c)

The source solutions state that the two curves do not intersect at the pole: C_1 reaches it at $\theta = 0$, while C_2 reaches it at $\theta = -\frac{\pi}{2}$.

40

The curves meet when $1 - \cos\theta = 1 + \cos\theta$, so $\cos\theta = 0$ and $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$. Thus

$$\begin{aligned} A_R &= \frac{1}{2}\int_{\pi/2}^{3\pi/2} \left((1-\cos\theta)^2 - (1+\cos\theta)^2\right) d\theta \\ &= -2\int_{\pi/2}^{3\pi/2} \cos\theta\, d\theta = [-2\sin\theta]_{\pi/2}^{3\pi/2} = 4 \text{ units}^2. \end{aligned}$$

41

Equating the curves and squaring gives $2\cos^2\theta\sin\theta = 3^{-1/2}\sin^2\theta$, so the non-pole intersection is at $\sin\theta = \frac{\sqrt{3}}{2}$, hence $\theta = \frac{\pi}{3}$. The smaller radius is r_2 for $0 \le \theta \le \frac{\pi}{3}$ and r_1 for $\frac{\pi}{3} \le \theta \le \frac{\pi}{2}$. Therefore

$$\begin{aligned} A_R &= \frac{1}{2}\int_0^{\pi/3} (3^{-1/4}\sin\theta)^2 d\theta + \frac{1}{2}\int_{\pi/3}^{\pi/2} (\cos\theta\sqrt{2\sin\theta})^2 d\theta \\ &= \frac{1}{2\sqrt{3}}\left(\frac{\pi}{6} - \frac{\sqrt{3}}{8}\right) + \left[-\frac{1}{3}\cos^3\theta\right]_{\pi/3}^{\pi/2} \\ &= \frac{\pi}{12\sqrt{3}} - \frac{1}{48} \text{ units}^2. \end{aligned}$$

42

The intersections satisfy $4\sin\theta = 3 - 2\sin\theta$, hence $\sin\theta = \frac{1}{2}$ and $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$. The common region is bounded by r_1 on $[0, \frac{\pi}{6}]$ and $[\frac{5\pi}{6}, \pi]$, and by r_2 between those angles. Thus

$$\begin{aligned} A_R &= \frac{1}{2}\int_0^{\pi/6} (4\sin\theta)^2 d\theta + \frac{1}{2}\int_{\pi/6}^{5\pi/6} (3 - 2\sin\theta)^2 d\theta \\ &\quad + \frac{1}{2}\int_{5\pi/6}^{\pi} (4\sin\theta)^2 d\theta \\ &= \left(\frac{2\pi}{3} - \sqrt{3}\right) + \left(\frac{11\pi}{3} - \frac{11\sqrt{3}}{2}\right) + \left(\frac{2\pi}{3} - \sqrt{3}\right) \\ &= 5\pi - \frac{15\sqrt{3}}{2} \text{ units}^2. \end{aligned}$$

43(a)

Equating the radii gives

$$\frac{7\sqrt{3}}{4\sin\theta} = 4 + \cos\theta.$$

For $\theta = \frac{2\pi}{3}$, both sides equal $\frac{7}{2}$. The other solution is $\theta \approx 0.68817$ radians, where the two radii are approximately equal.

43(b)

The line $r = \frac{7\sqrt{3}}{4}\operatorname{cosec}\theta$ is horizontal because $r\sin\theta = \frac{7\sqrt{3}}{4}$. Add this line to the diagram and mark the two intersections.

43(c)

Let $\alpha = 0.68817$. The required area is

$$A_R = \frac{1}{2}\int_0^{\alpha}(4+\cos\theta)^2 d\theta + \frac{1}{2}\int_{\alpha}^{2\pi/3}\left[\left(\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta\right)^2 - (4+\cos\theta)^2\right]d\theta + \frac{1}{2}\int_{2\pi/3}^{\pi}(4+\cos\theta)^2 d\theta.$$

Antiderivatives are

$$\frac{1}{2}\int(4+\cos\theta)^2 d\theta = \frac{33}{4}\theta + 4\sin\theta + \frac{1}{4}\sin 2\theta, \qquad \frac{1}{2}\int\left(\frac{7\sqrt{3}}{4}\operatorname{cosec}\theta\right)^2 d\theta = -\frac{147}{32}\cot\theta.$$

Substitution of the limits gives $A_R \approx 22$ units^2 to 2 significant figures.

44(a)

The curves intersect when $3 + 2\sin\theta = 2$, so $\sin\theta = -\frac{1}{2}$ and $\theta = -\frac{\pi}{6}, \frac{7\pi}{6}$. Hence

$$\begin{aligned} A_R &= \frac{1}{2}\int_{-\pi/6}^{7\pi/6}\left((3+2\sin\theta)^2 - 2^2\right)d\theta \\ &= \int_{-\pi/6}^{7\pi/6}\left(\frac{7}{2} + 6\sin\theta - \cos 2\theta\right)d\theta \\ &= \left[\frac{7}{2}\theta - 6\cos\theta - \frac{1}{2}\sin 2\theta\right]_{-\pi/6}^{7\pi/6} = \left(\frac{14\pi}{3} + \frac{11\sqrt{3}}{2}\right) \text{ units}^2. \end{aligned}$$

44(b)

The intersections satisfy $4 - 2\cos\theta = 6\cos\theta$, so $\cos\theta = \frac{1}{2}$ and $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$. The required area is

$$\begin{aligned} A_R &= \frac{1}{2}\int_{\pi/3}^{\pi}(4-2\cos\theta)^2 d\theta - \frac{1}{2}\int_{\pi/3}^{\pi/2}(6\cos\theta)^2 d\theta \\ &= \left[9\theta - 8\sin\theta + \frac{1}{2}\sin 2\theta\right]_{\pi/3}^{\pi} - \left[9\theta + \frac{9}{2}\sin 2\theta\right]_{\pi/3}^{\pi/2} \\ &= 6\sqrt{3} + \frac{9\pi}{2} \text{ units}^2. \end{aligned}$$

45(a)

Sketch the circle $r = 2$ on the same axes as the given curve. Its intersection with $r = 2\sinh\theta$ occurs when $\sinh\theta = 1$, so $\theta = \ln(1+\sqrt{2})$.

45(b)

The region is symmetric about $\theta = \frac{\pi}{2}$. Its half-area is

$$\begin{aligned} A_{1/2} &= \frac{1}{2}\int_0^{\ln(1+\sqrt{2})} (2\sinh\theta)^2 d\theta + \frac{1}{2}\int_{\ln(1+\sqrt{2})}^{\pi/2} 2^2 d\theta \\ &= \int_0^{\ln(1+\sqrt{2})} (\cosh 2\theta - 1) d\theta + \int_{\ln(1+\sqrt{2})}^{\pi/2} 2 d\theta \\ &= \sqrt{2} + \pi - 3\ln(1+\sqrt{2}). \end{aligned}$$

Doubling gives $A_R = 2\sqrt{2} + 2\pi - 6\ln(1+\sqrt{2})$ units^2.

46

The intersections satisfy

$$2\sin\theta = \frac{2\sqrt{3}}{3}(1-\cos\theta),$$

so $\sqrt{3}\sin\theta + \cos\theta = 1$. Writing the left side as $2\cos(\theta - \frac{\pi}{3})$ gives $\theta = 0$ or $\theta = \frac{2\pi}{3}$. The area is

$$\begin{aligned} A_R &= \frac{1}{2}\int_0^{2\pi/3} \left(\frac{2\sqrt{3}}{3}(1-\cos\theta)\right)^2 d\theta + \frac{1}{2}\int_{2\pi/3}^{\pi} (2\sin\theta)^2 d\theta \\ &= \left[\theta - \frac{4}{3}\sin\theta + \frac{1}{6}\sin 2\theta\right]_0^{2\pi/3} + \left[\theta - \frac{1}{2}\sin 2\theta\right]_{2\pi/3}^{\pi} \\ &= \pi - \sqrt{3} \text{ units}^2. \end{aligned}$$

47

The intersections satisfy $2\cos\theta = 1 + \sin\theta$, or $2\cos\theta - \sin\theta = 1$. Write $2\cos\theta - \sin\theta = \sqrt{5}\cos(\theta + \tan^{-1}\frac{1}{2})$. The acute solution is $\phi = \cos^{-1}\frac{1}{\sqrt{5}} - \tan^{-1}\frac{1}{2}$. From this, $\sin\phi = \frac{3}{5}$, $\cos\phi = \frac{4}{5}$, and $\sin 2\phi = \frac{24}{25}$.

$$\begin{aligned} A_R &= \frac{1}{2}\int_0^{\phi} (1+\sin\theta)^2 d\theta + \frac{1}{2}\int_{\phi}^{\pi/2} (2\cos\theta)^2 d\theta \\ &= \left[\frac{3}{4}\theta - \cos\theta - \frac{1}{8}\sin 2\theta\right]_0^{\phi} + \left[\theta + \frac{1}{2}\sin 2\theta\right]_{\phi}^{\pi/2} \\ &= \frac{\pi}{2} - \frac{2}{5} - \frac{1}{4}\phi \text{ units}^2. \end{aligned}$$

Thus $p = \frac{1}{2}$, $q = -\frac{2}{5}$, and $r = -\frac{1}{4}$.